Bishop-Pattern-Recognition-and-Machine-Learning-2006
PDF-dokument - 18,1 MB

0-Success-Stories-of-ML
PDF-dokument - 2,4 MB

01_Basic_Concepts
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exam format
MCQ 4 hrs
All assistive tools allowed

Algorithm whose performance increase with experience

$h = A(S, H)$

hypothesis $S = \begin{Bmatrix} (x_1, y_1) \\ \vdots \\ (x_m, y_m) \end{Bmatrix}$ input, label

$y = f(x)$ labeling function

$\boxed{\text{labeling function is unknown}}$

hypotosis space
which answers are allowed
notation: $\mathcal{H}$

for this
$f : \underbrace{X}_{\text{feature space}} \to \underbrace{Y}_{\text{label space}}$
(set of possible outputs)

for a function
example
$y(x) = \sum_{m=0}^{M} w_m x^m$

independence of input data
if $(x_1 \to y_1)$
then x_2 is unknown
i.i.d. = independent and identically distributed

w is a parameter (unknown)
M is a hyperparameter

	knowns	unknowns
task/Performance	S T/P X, Y	f $S \sim \underline{D}$ ← data

Loss function

0/1 loss $\quad \ell(y, y') = \mathbb{I}[y \neq y'] = \mathbb{I}[f(x) \neq h(x)]$

true label, Prediction

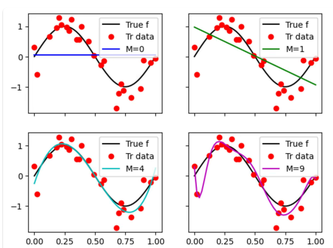
$L(h) = P_{X,Y}(\mathbb{I}[f(x) \neq h(x)])$

Score of success on all data

$\boxed{\text{Generalization Error}}$

we want to minimizes this

$h_* := \arg\min_h L_{(X,Y)}(h)$



We should "regularize"

$\arg\min_{h \in \mathcal{H}} L_{S_{train}}(h) + \lambda |\mathcal{H}|$

regularization coefficient

$L_S(h) = \frac{1}{m} \sqrt{\sum_{i=1}^{m} (h(x_i) - y_i)^2}$ $\quad$ T can be optional

root mean squared error

for

$\sum_{m=0}^{M} w_m \cdot x^m = (w_0 x^0 + w_1 x^1 \cdots w_m x^m)$

$= \vec{w} \cdot \vec{\phi} = \vec{w}^T \vec{\phi}$

where $\vec{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_m \end{bmatrix}$ $\quad \vec{\phi} \begin{bmatrix} 1 \\ x \\ \vdots \\ x^m \end{bmatrix}$

02_Linear_Predictors
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for $S = \{(x_i, y_i) : 1, 2, \cdots, m\}$

$f(x) = \vec{w}^T \vec{\phi}(x)$ $\quad$ where $\vec{\phi}(x) = \begin{bmatrix} 1 \\ x \\ \vdots \\ x^m \end{bmatrix}$

$\ell(y, y') = (y - y')^2$ $\quad$ and

$L_S(w) = \frac{1}{m} \sum_{i=1}^{m} (w^T \phi(x_i) - y_i)^2$ $\quad$ & $\vec{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_m \end{bmatrix}$

$A(S, H)$

$\boxed{w^* = \arg\min_w L_S(w)}$

$\frac{dL_S(w)}{dw} = 0$

$\Updownarrow$

$\nabla_{w^*} L_S(w^*) = 0$

for $p > 0$, the L_p-norm of a vector is defined as follows

$\|w\|_p = \left(\sum_{j=1}^{k} |w_j|^p\right)^{\frac{1}{p}}$

for

$L_S(h) + \lambda \cdot \|w\|_2^2 \to$ Ridge regression (you can find opt w)

$L_S(h) + \lambda \|w\|_2^1 \to$ Lasso regression (you have to use gradient decent) (or something else)

PyTorch: Autograd

Remember to optimizer.zero_grad()

autodiff-software
PDF-dokument - 6,4 MB

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Logistic Regression

note: $X = \{x_1, x_2\}$ $\quad \ell_{01}(y, y') = \mathbb{I}(y \neq y')$

$Y \in \{0, 1\}$ → Binary classification

$P(HT) = (1 - P_1) \cdot P_2$

$P(y = 1 | x_1) = P_1$
$P(y = 0 | x_2) = (1 - P_2)$

$P(y = a | x) = P_1^a \cdot (1 - P_2)^{1-a}$

Since $P(A, B) = P(A)P(B)$

$P(y_1 = 0, y_2 = 1 | x_1, x_2) = P(y_1 = 0 | x_1) P(y_2 = 1 | x_2)$

$= \prod_{i=1}^{2} P_i^{y_i} (1 - P_i)^{1 - y_i}$

$\mathcal{H} = \{h : h(x) = w^T x, \ w \in \mathbb{R}^k\}$
where k is dimentions

$\log\left(\frac{P_i}{1 - P_i}\right) = w^T x_i$

$\frac{P_i}{1 - P_i} = e^{w^T x_i}$

$P_i = e^{w^T x_i} \cdot (1 - P_i)$

$= e^{w^T x_i} - P_i \cdot e^{w^T x_i}$

$P_i \cdot e^{w^T x_i} + P_i = e^{w^T x_i}$

$P_i (e^{w^T x_i} + 1) = e^{w^T x_i}$

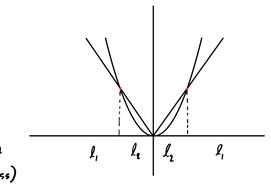
$P_i = \frac{e^{w^T x_i}}{(e^{w^T x_i} + 1)}$

$= \frac{e^{w^T x_i} \cdot e^{-w^T x_i}}{(e^{w^T x_i} + 1) \cdot e^{-w^T x_i}}$

$= \frac{e^{w^T x_i} \cdot e^{-w^T x_i}}{e^{w^T x_i} \cdot e^{-w^T x_i} + e^{-w^T x_i}} = \frac{1}{1 + e^{-w^T x_i}}$

Questions from exercise

why Loss² and not abs(loss)



l_2: use squared loss function
l_1 use absolute loss function

note: for this course l_2 is good enough

why $\lambda w^T w$ as a parameter
no free lunch theorem?

05_Statistical_Learning_Theory
HTML-fakt - 530 kB

Softmax

$P(y_i = c) = \frac{e^{w_c^T x}}{e^{w_c^T x} + e^{w_c^T x} + e^{w_c^T x} +}$